

ADITYA

Cloud Infrastructure · Backend Systems · Cybersecurity

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EDUCATION

B.Tech Computer Science — Vellore Institute of Technology, Chennai

Aug 2023 – May 2027 | CGPA: 8.71 / 10

Coursework: Operating Systems, Computer Networks, Cloud Computing, Cryptography, Distributed Systems, Software Engineering

EXPERIENCE

Research Intern — Cybersecurity & Software Assurance | SAG, DRDO · Ministry of Defence, Delhi

May 2026 – Jul 2026

- Contributed to cybersecurity research and software assurance activities for defence software systems under strict confidentiality protocols.
- Performed SAST, secure code review, static analysis, and Secure SDLC validation within classified defence development workflows.

AI / ML Project Intern | Kirat Communication Pvt. Ltd., Haryana

Jun 2025 – Jul 2025

- Architected Raspberry Pi conversational voice-AI system (Flask + Whisper STT + Ollama LLM + Gemini API); reduced end-to-end pipeline latency by 40% via audio buffer optimisation and inference request batching.
- Designed RESTful multi-stage NLP inference orchestration layer; delivered production-ready MVP independently within 6 weeks for early-stage startup validation.

Chairperson — Backend & Systems | Nexus Club, VIT Chennai (200+ members)

Apr 2025 – May 2026

- Owned end-to-end technical infrastructure and event logistics for NexusForum with 1,800+ attendees, industry speakers, and competitive programming tracks.
- Delivered Git/GitHub version-control workshops to 200+ students; built club website and internal automation tooling (TypeScript, Next.js, Supabase).

PROJECTS

RetailLens — E-Commerce Analytics Pipeline [GitHub](#) — Python, pandas, SQL, Streamlit, scikit-learn, matplotlib, seaborn

- Designed scalable ETL data pipeline on 500K+ row UK Online Retail dataset; SQL aggregation layer, RFM segmentation, and cohort retention analysis surfacing 34% repeat-buyer rate across 37 product categories.
- Deployed Streamlit dashboard with regional revenue breakdowns, time-series trend visualisation, and logistic regression churn model at 81% accuracy; EDA notebook published on GitHub.

Ubuntu Server — Browser-Based VDI System — Linux, QEMU/KVM, NGINX, VNC, noVNC, Tailscale, Netplan, Bash

- Provisioned headless Kali Linux VM using QEMU/KVM on Ubuntu host; bridged VNC via Websockify + noVNC on port 6080; configured NGINX reverse proxy and static IP via Netplan for stable LAN access.
- Implemented layered authentication (VNC password + OS credentials) and WAN extension via Tailscale overlay networking for secure remote access.

CloudClip — Serverless Cross-Device Clipboard — AWS Lambda, API Gateway, Amplify, S3, JavaScript

- Serverless clipboard sync on AWS (Lambda + API Gateway + Amplify); sub-100ms API response measured via Postman; zero-infrastructure auto-scaling cloud-native architecture.

Fake News Classifier — Python, NLP, scikit-learn, pandas, TF-IDF

- NLP classification pipeline with TF-IDF vectorisation; benchmarked Naive Bayes, Logistic Regression, and SVM — achieved 94% test accuracy; precision/recall tradeoffs documented.

Certificate Automation System — Python, SMTP, PDF Generation

- Batch-generated and auto-emailed customised certificates to 500+ recipients via SMTP with delivery confirmation; reduced manual distribution time by 95%.

TECHNICAL SKILLS

Cloud & Infra: AWS (Lambda, API Gateway, Amplify, S3), Docker, Docker Compose, NGINX, CI/CD (GitHub Actions), QEMU/KVM, Virtualization, Linux SysAdmin, Netplan, Ubuntu Server, SSH, Bash/Shell, ngrok

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, Bash

Cybersecurity: SAST, Secure SDLC, Static Analysis, Secure Code Review, Cryptographic Protocol Implementation, OWASP Top 10, Kali Linux, Linux Hardening

Backend & APIs: FastAPI, Flask, Node.js, RESTful API Design, PostgreSQL, Supabase, Firebase, Microservices, Containerisation

Data & ML: pandas, NumPy, scikit-learn, matplotlib, seaborn, Streamlit, ETL Pipelines, EDA, TF-IDF, Jupyter Notebook

AI / LLMs & Frontend: Ollama, Gemini API, Whisper STT/TTS, LLM Integration, ML Model Deployment | React, Next.js, Tailwind CSS

Tools: Git, GitHub, Postman, VS Code, Raspberry Pi

CERTIFICATIONS & ACHIEVEMENTS

- NVIDIA Deep Learning Institute — Fundamentals of Deep Learning (2025)
- Defy '25 Aptos Side Track Winner — blockchain smart-contract solution recognised at national hackathon
- Chairperson, Nexus Club VIT Chennai — technical leadership for 200+ member community; NexusForum with 1,800+ attendees
- Open Source — 40+ public repositories spanning backend engineering, cloud infrastructure, and Linux systems; Pull Shark & Pair Extraordinaire GitHub achievements
- Competitive Programming — problem setter, mentor, and technical judge across 5+ hackathons and contests, CodeChef VIT-Chennai Chapter